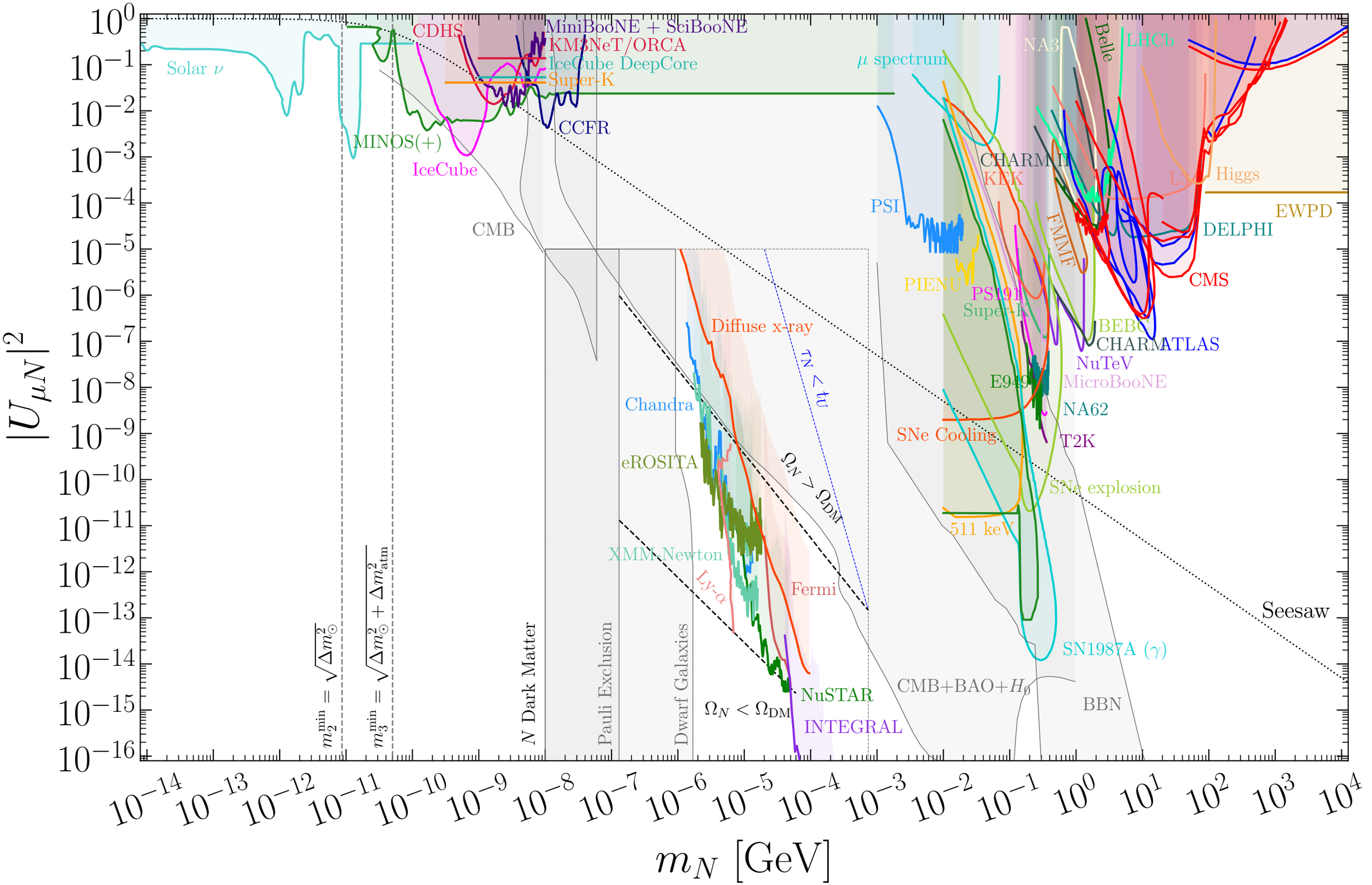

$|U_{\mu N}|^2$
m_N [GeV]
Solar ν
CDHS
MiniBooNE + SciBooNE
KM3NeT/ORCA
IceCube DeepCore
Super-K
CCFR
MINOS(+)
IceCube
CMB
μ spectrum
NA3
Belle
LHCb
CHARM II
KEK
L3
Higgs
EWPD
PSI
FMMF
DELPHI
CMS
PIENU
PS191
Super-K
BEBC
CHARM
ATLAS
NuTeV
MicroBooNE
E949
NA62
T2K
SNe Cooling
SNe explosion
511 keV
SN1987A (γ)
CMB+BAO+H_0
BBN
Seesaw
Diffuse x-ray
$\tau_N < t_U$
Chandra
eROSITA
$\Omega_N > \Omega_{\rm DM}$
XMM-Newton
Ly-α
Fermi
NuSTAR
$\Omega_N < \Omega_{\rm DM}$
INTEGRAL
$m_2^{\rm min} = \sqrt{\Delta m_\odot^2}$
$m_3^{\rm min} = \sqrt{\Delta m_\odot^2 + \Delta m_{\rm atm}^2}$
N Dark Matter
Pauli Exclusion
Dwarf Galaxies